

# Model Evaluation

Goodness of Fit for Logistic and Poisson

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## Last week: Model Evaluation for MLR

Does the model fit the data well? Do we need more/less variables in the model?

- Goodness of fit  $R^2$  (adjusted  $R^2$ ), RSE, MSE
- $F$ -test to compare our model against an intercept-only model
- a more general  $F$ -test to compare nested models

Most metrics used to evaluate LR measure how far  $y$  is from  $\hat{y}$   
Can we do the same for other logistic and Poisson models?

No

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Multiple linear regression models for continuous responses.  
normal  
(functions of  $y_i - \hat{y}_i$ )

## Today: Model Evaluation beyond MLR

- Residuals
- Deviance for GLM
- Chi-square test

→ for non-normal responses

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## Prediction: Logistic Regression for binary responses

An estimated logistic model can be used to predict an outcome

- the predicted outcome depends on the form of the model
  - log-odds (default, since this is a linear model)
  - probabilities
  - odds
- we can predict the outcome of:
  - observations in the same sample used to estimate the model, called fitted values  $\hat{\eta}_i$
  - new observations, usually referred as predicted values  $\hat{\eta}_i$

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## Prediction: Poisson Regression

An estimated Poisson model can be used to predict an outcome

- the predicted outcome depends on the form of the model
  - log-mean (default, since this is a linear model)
  - mean
- we can predict the outcome of:
  - observations in the same sample used to estimate the model, called fitted values  $\hat{y}_i$
  - new observations, usually referred as predicted values  $\hat{y}_i$

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## Residuals for Logistic regression

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The **raw residuals** are, as usual, the differences between the observed and the fitted values.

$$\begin{aligned} r_i &= \text{obs}_i - \text{fit}_i \\ &= y_i - \hat{p}_i \end{aligned}$$

However, there are 2 problems with these residuals:

- the variance of the observations is not constant! for a binary response:

$$\text{Var}(Y_i) = p_i(1 - p_i) = E(y_i)(1 - E(y_i))$$

technically, we should write the conditional variance and probability

Thus, residuals of different observations are not comparable and should be adjusted!!

$$\begin{aligned} \log\left(\frac{E(y_i)}{1 - E(y_i)}\right) &= \beta_0 + \beta_1 x_i \\ E(y_i) &= \frac{e^{\beta_0 + \beta_1 x_i}}{1 + e^{\beta_0 + \beta_1 x_i}} \end{aligned}$$

$$E(y_i | x_i)$$

$$\begin{aligned} \text{Linear model} \\ E(y_i) &= \beta_0 + \beta_1 x_i \\ \text{Var}(y_i) &= \sigma^2 \\ &\text{(unrelated to } E(y_i)) \end{aligned}$$

Many variations of the raw residuals have been proposed for Logistic regression (and GLMs in general) that take the variance of the observations into account.

- **Pearson residuals:** divides the raw residuals by the standard deviation of the response

$$r_i = \frac{y_i - \hat{p}_i}{\sqrt{\hat{p}_i(1 - \hat{p}_i)}} = \frac{y_i - \hat{E}(y_i)}{\sqrt{\text{Var}(y_i)}}$$

- deviance residuals
- **standardized residuals:** for both Pearson and deviance residual

We won't get to the details of these but it is important to know which ones you are calculating.

## Residuals for Poisson regression

The Poisson regression has the same problems as the Logistic Regression:

- responses are discrete (at least more than 2 values now)
- the variance is not constant

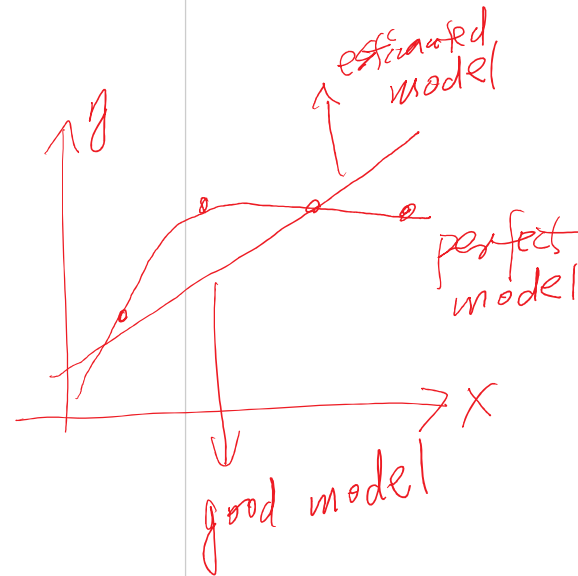
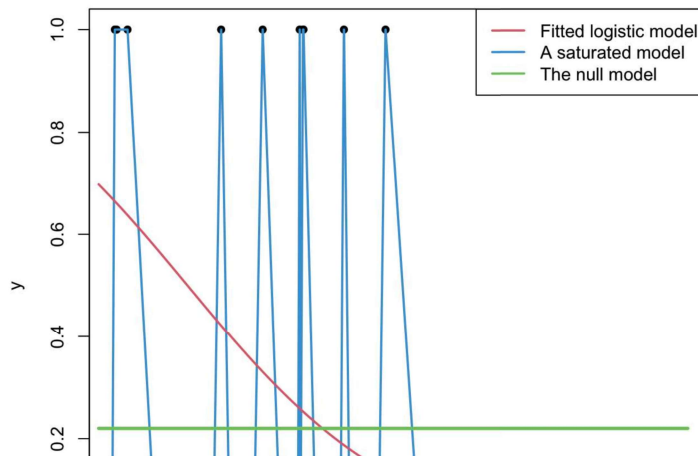
$$\text{resid}_{\text{raw}} = y_i - \hat{\lambda}_i, \quad \text{resid}_{\text{pearson}} = \frac{y_i - \hat{\lambda}_i}{\sqrt{\hat{\lambda}_i}} = \frac{y_i - \hat{E}(y_i)}{\sqrt{\text{Var}(\hat{E}(y_i))}}$$

↑  
not useful

# Deviance

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Beyond MLR, a quantity called the deviance is used to measure the difference between the log-likelihoods the estimated model and that of a perfect model, known as the saturated model, which fits the data perfectly.



## Deviance: interpretation

The deviance generalizes the Residual Sum of Squares (RSS) of the linear model.

If the errors of a LR are normal, the deviance is proportional to the RSS.

The lower the deviance the better the fit!

$$R^2 = 1 - \frac{RSS}{TSS}$$

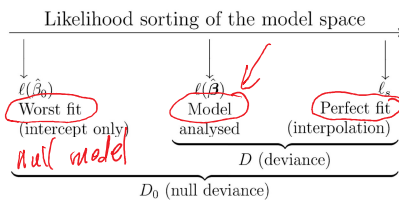
Mathematical statistics  $\rightarrow$  statistics  
 $\rightarrow$  statistical sciences  $\rightarrow$  data science

deviance = RSS for linear models

$$RSS = \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

$$= \sum_{i=1}^n (\text{residuals})^2$$

## Deviance to test



$H_0$ : models I and II are equally good  
 $H_1$ : one model is better

Image from [Notes for Predictive Modeling](#)

The deviance can also be used to compare nested models for GLM (similar to the  $F$ -test in LR).

The test that compares deviances has a  $\chi^2$  distribution.

Deviance  $\sim \chi^2(d)$ , where  $d$  is the difference of the number of predictors in the two models.

test statistic  $\rightarrow$  under  $H_0$

Deviance measures the difference of the two log-likelihoods of the two models

Only works well for large samples

$n \rightarrow \infty$   
 asymptotics

Don't worry, R computes this test for you using `anova()`!

logistic models  
→ anova(model I, model II)